

Level 1 — 100 marks/question**L1-Q1. (E) Prime-Composite**

P

CODE

```
import math

def is_prime(n: int) → bool:
    if n ≤ 1:
        return False
    if n == 2:
        return True
    if n % 2 == 0:
        return False
    r = int(math.isqrt(n))
    for i in range(3, r + 1, 2):
        if n % i == 0:
            return False
    return True

numbers = []
count = 0

while True:
    try:
        line = input().strip()
        if line == "":
            continue
        numbers.append(int(float(line)))
        count += 1
    except EOFError:
        break

prime_count = 0
```

```
composite_count = 0

for n in numbers:
    if n > 1 and is_prime(n):
        prime_count += 1
    elif n > 1 and not is_prime(n):
        composite_count += 1

print(f"Composite number:{composite_count}")
print(f"Prime number:{prime_count}")
```

INPUT

```
4
54
29
71
7
59
98
23
```

OUTPUT

```
Composite number:3
Prime number:5
```

L1-Q2. (E) Duplicated 1D

P

CODE

```
def remove_duplicates(arr):
    seen = set()
    result = []
    for x in arr:
        if x not in seen:
            seen.add(x)
            result.append(x)
    return result

arr = list(map(float, input().strip().strip('{}').split(',')))

if all(x.is_integer() for x in arr):
    arr = [int(x) for x in arr]

ans = remove_duplicates(arr)
print("{ " + ",".join(map(lambda v: str(int(v)) if isinstance(v, float) and v.is_integer()
```

INPUT

{12,23,45,45,67,88,12}

OUTPUT

{12,23,45,67,88}

L1-Q3. (E) Vowels

P

CODE

```
s = input().strip().lower()

vowels = set('aeiou')
vowel_count = 0
consonant_count = 0

for ch in s:
    if ch.isalpha():
        if ch in vowels:
            vowel_count += 1
```

```
if ch in vowels:
    vowel_count += 1
else:
    consonant_count += 1

print(f"Number of vowels : {vowel_count}")
print(f"Number of consonant : {consonant_count}")
```

INPUT

string

OUTPUT

Number of vowels : 1
Number of consonant : 5

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L1-Q4. (E) RT-Pattern

P

CODE

```
ch = input("Enter the Character to be printed: ")
n = int(input("Number of rows.: "))

for i in range(1, n + 1):
    print((ch + " ") * i, end="")
    print()
```

INPUT

+

5

OUTPUT

```
Enter the Character to be printed: Number of rows.: +
+ +
+ + +
+ + + +
+ + + + +
```